THE STALK OF THE NORMALIZATION AT A POINT IS THE INTEGRAL CLOSURE IN THE PRODUCT OF RESIDUE FIELDS OF THE MINIMAL PRIMES OF THE LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 14465-14479 and proof span 14481-14501 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof offers two alternative routes to the first identification (the construction of the normalization, or the description of the normalization with localization of integral closures); both are preserved as the author wrote them.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The construction of the normalization and the identification of its pushforward of the structure sheaf by integral closures, cited from the same chapter. Cited at proof lines 14481-14485. Gives the identification of the stalk of the pushforward with the integral closure in a product of residue fields.
- An alternative description of the normalization, and the compatibility of integral closures with localization, cited from the same chapter and the algebra chapter. Cited at proof lines 14485-14488. Supplies a second route to the same identification, so that the two descriptions are seen to agree.
- For a reduced ring with finitely many minimal primes the product of the residue fields at those primes is the total ring of fractions. Cited at proof lines 14492-14495. Identifies the product appearing in the second description with the total ring of fractions of the reduced local ring.

Source reference. Original lemma 0307: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3938–3938. Same-chapter reference used in the retained author text at source line(s) 13874, 14339, 14493. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers

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Source reference. Original lemma 02LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3424–3424. Same-chapter reference used in the retained author text at source line(s) 14451, 14498. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 035P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4136–4136. Same-chapter reference used in the retained author text at source line(s) 14491, 14535, 14570, 14599, 14678, 14700, 15083. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 035F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4126–4126. Same-chapter reference used in the retained author text at source line(s) 13884, 13969, 14199, 14489. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a scheme such that every quasi-compact open has a finite number of irreducible components. Let $\nu : X^\nu \to X$ be the normalization of X. Let $x \in X$. Then the following are canonically isomorphic as $\mathcal{O}_{X,x}$-algebras*

(1) *the stalk $(\nu_*\mathcal{O}_{X^\nu})_x$,*
(2) *the integral closure of $\mathcal{O}_{X,x}$ in the total ring of fractions of $(\mathcal{O}_{X,x})_{red}$,*
(3) *the integral closure of $\mathcal{O}_{X,x}$ in the product of the residue fields of the minimal primes of $\mathcal{O}_{X,x}$ (and there are finitely many of these).*

Proof. After replacing X by an affine open neighbourhood of x we may assume that X has finitely many irreducible components and that x is contained in each of them. Then the stalk $(\nu_*\mathcal{O}_{X^\nu})_x$ is the integral closure of $A = \mathcal{O}_{X,x}$ in the product L of the residue fields of the minimal primes of A. This follows from the construction of the normalization and Lemma 035F. Alternatively, you can use Lemma 035P and the fact that normalization commutes with localization (Algebra, Lemma 0307). Since A_{red} has finitely many minimal primes (because these correspond exactly to the generic points of the irreducible components of X passing through x) we see that L is the total ring of fractions of A_{red} (Algebra, Lemma 02LX). Thus our ring is also the integral closure of A in the total ring of fractions of A_{red}. □